

Element K

The Nashville Zoo needed an improved animal lift to safely transport injured or sedated animals without causing additional harm. Our primary goals were to design a lift with a weight-bearing capacity of 50-1100 pounds, balance the weight evenly across its center of mass, ensure durability, and facilitate ease of use and removability. Reflecting on our brainstorming sessions, we considered various materials, shapes, and mechanisms, ultimately settling on a sturdy metal frame with adjustable straps to support various animal sizes.

We selected unistrut/superstrut metal for the frame due to its strength and durability. The design included a rectangular frame with corner braces and a balance mechanism. Our aim was to create a prototype that connected easily to existing mats using straps and carabiners, ensuring stability and safety during lifting. The construction process was both challenging and enlightening. We cut unistrut into specific lengths (4ft by 6ft), assembled the frame with corner brackets, and attached straps with carabiners, reusing materials from previous projects to minimize costs and waste. During testing, we focused on weight capacity and balance. Hanging the frame from the ceiling and tested the engine lift to determine how well it balanced the various loads. Adjustments to improve stability were necessary, providing valuable insights into the design and construction processes. However, despite these efforts the final prototype did not fully meet all design criteria. The connections were not as efficient as planned, affecting the weight requirement and stability. Nevertheless, the project offered important learning experiences.

Several aspects of the project went well. Choosing unistrut metal for the frame provided the necessary strength the old frame was lacking. Reusing nuts, bolts, and straps from previous projects reduced costs and promoted sustainability. Iterative changes improved the prototype's

overall performance, addressing issues like torque and balance. Not everything went smoothly, the initial method for connecting the frame sections was inefficient, leading to stability issues. The final prototype struggled to meet the desired weight capacity due to connection inefficiencies and time and material limitations. Additionally, the assembly process was more complex and time-consuming than anticipated, impacting ease of use.

From this project, we learned several key lessons. Such as in the design process, prototyping is crucial for identifying and solving design flaws. Early testing can save time and resources by highlighting issues before final construction. Selecting the right materials is vital, considering strength, durability, and ease of assembly. Effective connection methods are essential for maintaining stability and meeting design requirements. Future designs should prioritize and prototype efficient connections early. Flexibility and openness to changes throughout the design process can lead to better outcomes because unexpected challenges require adaptive solutions. Allocating sufficient time for each phase of the project, especially testing and adjustments, is critical for successful completion. Keeping detailed records of each step, including design changes and testing results, aids in troubleshooting and improves communication within the team.

Thorough planning is necessary for a clear understanding of the problem and design requirements. Detailed planning can prevent many issues during the construction phase. Regular testing and iterative improvements are key to developing a functional and reliable product. Effective teamwork and communication are essential, as drawing on diverse skills and perspectives can enhance the design process. Considering the environmental impact of materials and processes is also important. Reusing materials and tools can reduce costs and promote

sustainability. By reflecting on our experience with the animal lift project, we can provide valuable insights into future engineering projects.